

Evaluating User Experience Across Platforms: Analyzing Complex Task Performance in Academic Desktop-Mobile Platforms

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Abstract— User experience varies widely when using the desktop and mobile versions of an application. This difference is obvious in several aspects, including usability, performance, functionality, user preferences, and accessibility. This study compares the usability outcomes of ResearchGate, a widely used academic collaboration platform, on both desktop and mobile devices. Ten participants made up the sample, and they received a comprehensive scenario. The scenario was divided into five tasks that must be accomplished in a specific order. According to the results of this study, there was a significant variation between the desktop and mobile versions, with the desktop version showing more efficient performance regarding usability and overall experience. These findings highlight critical design considerations for academic platforms aiming to optimize cross-device usability. Additionally, this study contributes to broader usability research by providing empirical evidence on how interface disparities affect user behavior, offering actionable insights for developers prioritizing seamless multi-device experiences.

Keywords—Usability evaluation; user experience; software engineering

I. INTRODUCTION

Nowadays, software is prevalent and plays a vital role in every aspect of our daily lives. It is becoming increasingly important in almost every business domain, like education, finance, communication, etc. As well as providing safety-critical functions in medical, transportation, and nuclear energy fields, software is one of the most challenging and leading technologies of our time. Moreover, it is among the most complicated and error-prone technologies. Therefore, the quality of the software is paramount and plays a primary role [1].

Since the beginning of programming, developers and users have been concerned with quality issues because poor software quality could lead to software failure, which can have significant consequences, including customer dissatisfaction, loss of financial assets, and even loss of human lives. Thus, software quality is an area of extreme importance in today's marketplace [1].

The quality of software is one of the most elusive concepts. It comes in many forms and can be expressed in different aspects. Since the 1970s, significant efforts have been made to define software quality to help understand, evaluate, and improve it [2].

Generally, quality means "the ability of the product to perform satisfactorily and be suitable for its intended

purpose." The ISO 14598-1 standard defines quality as "the totality of characteristics and features of a product or service that bear on its ability to satisfy stated and implied needs." The IEEE 610.12 standard, on the other hand, defines software quality as "the degree to which a system, component, or process meets specified requirements and customer/user needs or expectations" [3].

Quality comprises various characteristics, including usability, security, and functionality. Thus, the quality of the software is studied and evaluated using specifically constructed quality models. The purpose of these models is to define the fundamental quality characteristics. Each model encompasses multiple sub-characteristics, and metrics are assigned to each sub-characteristic for the actual assessment. [4], [5].

In discussions regarding quality characteristics, usability is the first feature noticed by users when interacting with any software. It is also one of the essential quality characteristics that plays a significant role and supports the overall quality of all software types.

In software engineering literature, several experts have provided a definition of the usability concept. In general, usability means "the simplicity of learnability, remembrance, and use of the software by end users." From a technical perspective, usability can be defined as "the degree to which a software product is employed by particular groups to achieve the goals of efficiency, effectiveness, satisfaction, and many other features." [6]. In addition, the ISO/IEC 9126-1 standard defined usability as "the capability of the software product to be understood, learned, used, and attractive to the user when used under specified conditions." [7]. The IEEE standard 610.12-1990, on the other hand, described usability as "the ease with which a user can learn to operate, prepare inputs for, and interpret outputs of a system or component." [8].

II. LITERATURE REVIEW

Most previous studies that explored the comparison between desktop and mobile versions of the same

application focus on a complex interplay between usability, accessibility, and performance, each of which has its strengths and weaknesses. According to Smith [9], mobile apps excel in scenarios that require quick interactions on the go, such as browsing messages and social media apps. Moreover, mobile apps have simplicity. However, this simplicity often comes at the expense of advanced features, as mobile apps are typically designed to fit the small screens and limited hardware capabilities of these devices. On the other hand, desktop applications are designed for quality and precision, making them the preferred choice for complicated tasks such as graphic editing of videos, software development, and data analysis. With their large screens, powerful hardware capabilities, and multitasking ability, desktop apps provide a high level of performance and customization that is considered challenging for mobile apps to match [10].

Usability studies confirm this difference, showing that desktop applications consistently exceed their mobile equivalents regarding resource efficiency, speed, and multitasking capabilities. An example of this difference can be seen in Adobe Premiere Pro, a desktop software that renders video significantly faster than its mobile counterparts, which usually rely on cloud processing to compensate for hardware limitations [11]. Despite these challenges, mobile applications have witnessed significant improvements in terms of performance using advanced techniques such as data compression, efficient memory management, and leveraging cloud services, which facilitate their ability to handle more difficult tasks [12].

Functionality is another remarkable difference when comparing desktop and mobile versions. Desktop applications typically have a wide range of advanced tools and characteristics, making them the ideal option for professional users who need superior capabilities. For example, the desktop version of Microsoft Office offers complete access to macros, add-ons, and advanced formatting options. In contrast, the mobile version focuses on basic tasks such as document editing and collaboration, sacrificing some advanced features for ease of use [13]. Likewise, Adobe Photoshop on the desktop provides a full range of advanced editing tools. In addition, the mobile version, Photoshop Express, offers a specific collection of these tools that matches the quick editing on the go.

Another challenge faced in this landscape is understanding user preferences, which are significantly impacted by the type of task required and users' demographic backgrounds. Research indicates that younger users, particularly millennials and Gen Z, prefer to use mobile apps to accomplish tasks that were previously performed on desktop computers, such as document editing and content creation; the reason for this is the greater familiarity of the young users and the convenience of mobile devices [14]. However, desktop applications are still the preferred choice for professional uses where tasks require high precision, and the need for advanced features and precise control is more significant than the need for portability.

Accessibility is another crucial part of this comparison. Mobile applications excel at integrating internal accessibility features like screen readers, voice

commands, and magnification tools. These features are seamlessly integrated into mobile operating systems, making them more user-friendly for people with disabilities [15]. On the other hand, desktop applications provide greater adaptability and customization in accessibility settings, allowing users to customize the interface according to their specific needs. For instance, desktop software such as JAWS (Job Access with Speech) offers a wide range of customization options for users with impaired vision, though the setup required is more than that of mobile solutions [16].

The complex challenges developers face with cross-platform application development include balancing between maintaining consistency across different platforms and achieving the unique requirements of each. React Native and Flutter are frameworks that have emerged as innovative solutions, allowing developers to build applications that can perform on desktop and mobile devices using a single code base. Nonetheless, these frameworks often require some compromises concerning performance and user experience quality, as they struggle to utilize the full and distinct capabilities of each platform [17]. Moreover, various input methods, like touch-based interactions on mobile and keyboard and mouse use on desktop, require a different design approach for each platform, which adds another complication to the development process and more challenges for developers [18].

These challenges make the development of innovative solutions that bridge the gap between desktop and mobile applications an urgent need, guaranteeing a smooth and fulfilling experience for users across all platforms. Future studies should focus on bridging these gaps, especially prioritizing areas like boosting performance, enhancing accessibility, and developing integrated cross-platform solutions to design applications that combine the best characteristics of each platform. By understanding and acting toward fulfilling the distinct requirements of each platform, developers can design applications that meet the diversity of users' needs, such as improving productivity, facilitating accessibility, and achieving overall user satisfaction.

III. METHODOLOGY

This research applied summative usability testing as the primary tool for evaluating product performance. This method relies on engaging real users in performing representative tasks specifically designed to measure the overall usability of the product through three main dimensions: effectiveness (the extent to which goals are achieved), efficiency (time and effort spent), and satisfaction (positive user experience). This type of testing aims to evaluate how easy it is to use an interactive system and how enjoyable the experience is for users. In short, summative usability testing is an effective tool for comprehensively analyzing the performance of a product and identifying strengths and weaknesses in the user experience.

A. Performance measurements

A summative test was used to collect quantitative data on participants' performance while performing specific

tasks in a usability test. This test was conducted in a formal laboratory to ensure the accuracy of the results and minimize interference. Before the test started, participants received training and were given a list of tasks. Throughout the experiment, participants were closely monitored to record their interactions and performance, ensuring reliable results regarding user effectiveness, efficiency, and satisfaction.

B. Procedure

The goals for the usability testing were defined in terms of usability attributes (e.g., easy to learn, efficient to use, easy to remember, few errors, subjectively pleasing). The various components of the goals were balanced, and their relative importance was decided. Usability issues were quantified by measurements such as those given below.

- The time users take to complete a specific task.
- The time spent recovering from errors.
- The number of user errors

C. Scope

This research study compares the usability of two different devices in terms of user efficiency. Specifically, it focuses on ResearchGate, a widely used academic collaboration platform that is accessible via both desktop website and mobile application. This platform was chosen due to its relevance to the researchers and students.

D. Equipment

High-quality laptops featuring robust operating systems and browsers, along with mobile phones that have excellent specifications.

E. Location

The test was held in a test laboratory at University of Benghazi.

F. Participants

Ten participants were selected for the usability test, which is ideal for identifying most issues according to Jakob Nielsen's methodology. This group consisted of researchers and graduate students who were familiar with using academic platforms, with a variety of academic backgrounds (humanities, natural sciences, engineering, etc.). This diversity ensured a broad representation of user needs.

G. Tasks

Participants were given a comprehensive scenario. It was divided into five tasks to be completed in a specific order. These tasks represent the typical workflow when using the ResearchGate platform. Table I shows the tasks assigned to all participants.

TABLE I. TASKS

No	Task Name
1	Registration
2	Profile Management
3	Publication Upload
4	Network Building
5	Research Collaboration

IV. RESULTS AND DISCUSSION

The evaluation revealed that users' performance on the desktop was more efficient and easier compared to the mobile platform. Users encountered fewer difficulties when completing tasks on the desktop. However, clear challenges were observed on the mobile platform, particularly with tasks requiring complex interactions, such as uploading posts and managing the profile. These tasks not only took longer to complete but also resulted in a higher number of errors. Furthermore, the network-building task was relatively complex on both platforms, indicating a need to simplify this functionality. Overall, the results suggest a need to enhance the user experience on the mobile platform, focusing on simplifying the design and reducing the steps required to complete key tasks.

A. Number of Mistakes

The analysis of the total number of errors revealed a notable increase in errors on the mobile platform compared to the desktop across all tasks. As shown in Table II, the highest error rates on mobile were observed in specific tasks, such as uploading publications, with four errors, and research collaboration, with seven errors. In comparison, tasks such as network building were more prone to errors on both platforms, with seven errors on desktop and nine on mobile. Although the number of errors in the registration task was equal (two errors per platform), the high number of errors on mobile in complex tasks indicates a pressing need to improve interface clarity and reduce design complexity, especially for tasks that require multi-step interactions. These findings emphasize the need to streamline the workflow on mobile to enhance the user experience and reduce the gap between the two platforms. Figure 1 shows the number of mistakes in each platform.

TABLE II. TOTAL TASK MISTAKES

No	Total Task Mistakes (num)	
	Desktop	Mobile
1	2	2
2	3	4
3	2	4
4	7	9
5	4	7



Figure 1. Number of mistakes in each task

B. Total Task and Mistake Time

Table III shows that the total time taken to complete tasks was significantly lower on desktop compared to mobile. Participants spent notably more time on mobile, particularly for tasks such as uploading posts, which took an average of 8 minutes on mobile, while it only took 3 minutes on desktop. Similarly, network building took 15 minutes on mobile and 10 minutes on desktop. Additionally, the amount of missed time during tasks was observed to be higher on mobile for most tasks, peaking in uploading posts, where mobile users spent an average of 9 minutes, compared to just 4 minutes on desktops. Similarly, for research collaboration, mobile users took 12 minutes, while desktop users took 7 minutes, as shown in Figure 2 and 3. These differences suggest that using mobile requires more effort from users to complete tasks successfully. Therefore, there is a clear need to improve the user experience on mobile, especially for complex tasks that require multi-step interactions.

TABLE III. TOTAL TASKS AND MISTAKE TIME

No	Total Task Time (min)		Mistake Time (min)	
	Desktop	Mobile	Desktop	Mobile
1	5.14	7.20	4.5	3
2	6	9	2.20	8
3	3	8	4	9
4	10	15	5	6
5	8.40	12	7	12

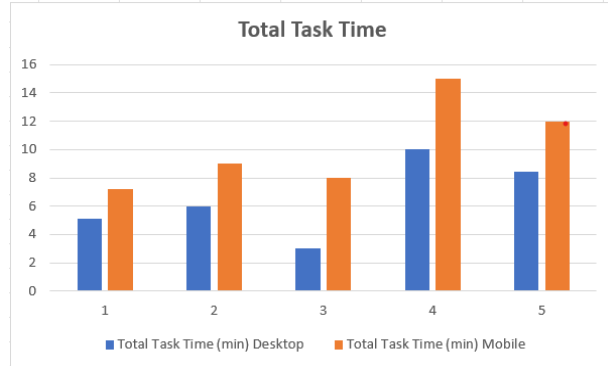


Figure 2. Comparison between the overall task time



Figure 3. Comparison between the overall mistake time.

V. CONCLUSION

A comparative study was conducted between desktop and mobile computers to evaluate usability. Ten researchers and graduate students familiar with academic platforms participated. The results showed that desktop computers were significantly superior to mobile computers in completing tasks faster and with fewer errors. For example, when uploading publications, users faced significant difficulties on mobile, taking longer and making more errors than on desktop. Similar challenges were observed in the research collaboration task, where the computer demonstrated clear superiority. The study revealed that mobile devices struggle with tasks that require multi-step interactions, such as locating storage or managing networks, which resulted in increased total time taken and more time spent on error. These findings indicate that the user experience on mobile needs substantial improvements. Key areas for enhancement include simplifying the design, improving clarity, organizing interactive elements, reducing the number of steps required, and providing better guidance. Overall, the results emphasize the importance of improving the user experience on mobile to ensure higher efficiency and a more satisfying user experience.

For future work, it is recommended to extend this study to include a larger sample and evaluate usability on other platforms.

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